

Mingze Liu

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EDUCATION

University of British Columbia (UBC) - Bachelor of Science, Major in Computer Science

Expected May 2029

Relevant coursework: Software Construction (CPSC 210), Models of Computation (CPSC 121), Systematic Program Design (CPSC 110), Data Science (DSCI 100)

OPEN-SOURCE EXPERIENCE

AIRI - Open-Source Contributor [GitHub](#)

Apr 2026 - Aug 2026

- Authored and merged 14 pull requests across a large TypeScript/Vue desktop application, spanning product features, reliability, security, documentation, and CI.
- Built desktop chat management and 3D character interactions; improved Linux runtime behavior and added regression tests for UI state and audio-pipeline edge cases.

PROJECTS

Desktop Personal Finance Tracker

Feb 2026 - Mar 2026

Java, Swing, JSON, JUnit

- Built an event-driven desktop application with account and transaction workflows, JSON persistence, modular object-oriented design, and JUnit coverage for edge cases.

Dynamic Audio Generation & Speech Server

Apr 2026

Python, Flask, REST APIs

- Developed REST endpoints for speech synthesis and voice indexing, with structured request validation and dynamic asset lookup for reliable audio delivery.

Windows on ARM Firmware - POCO F4 / Redmi K40S [GitHub](#)

Aug 2026

C, EDK2/UEFI, ARM64, GitHub Actions

- Used an AI-assisted development workflow to adapt EDK2 firmware for a Snapdragon 870 phone, iterating on memory maps, framebuffer initialization, NTFS support, and Windows boot handling.
- Tested builds on physical hardware, diagnosed boot and memory failures, validated Windows 11 ARM64 startup, and documented a reproducible installation and troubleshooting process.

TECHNICAL SKILLS

Languages: TypeScript, Java, Python, Go, C, R, Scheme/Racket, Bash

Frameworks & testing: Vue, Flask, Java Swing, JUnit, PyTest, REST APIs

Developer tools & systems: Git, GitHub, Docker, Linux (Ubuntu/Fedora), SSH, Tailscale, GCP, Azure, Hetzner Cloud, Raspberry Pi, GPIO/I2C